PH3120 – Computational Physics Laboratory 1

CPL104 – Vectors and Matrices

Week 05

In [2]:
```
# Q 1) (a)
```

In [27]:
```
import numpy as np
k = 8.988e9  # 1 / (4 * pi * epsilon)
charges = np.array([2e-6, -4e-6, 6e-6])
positions = np.array([[1, 0],[0, 2], [-2, -1]])
Electric_filed = np.array([0.0, 0.0])
print("Electric fileds :")
for i in range (0,3):
    q = charges[i]
    r = np.linalg.norm(positions[i])  # distance form charge to origin or magnitude
    vector = -positions[i]
    unit_vector= vector/r
    E= ((k*q)/(r**2))*unit_vector
    print (E)
    Electric_filed += E
print("\n")
print("Total Electric filed of origin : \n",Electric_filed)
print("\n")
magnitude = np.linalg.norm(Electric_filed)
angle = np.degrees(np.arctan2(Electric_filed[1], Electric_filed[0]))  # angle from

print(" Magnitude of Total Electric Field :\n ",magnitude)
print("\n")
print(" Direction (angle from x-axis in degrees):\n",angle)
print("\n")
unit_vector = Electric_filed / magnitude
print("Unit Vector of Total Electric Field:\n", unit_vector)
```

```
Electric fileds :
[-17976.      0.]
[  -0. 8988.]
[9646.93391125 4823.46695562]


Total Electric filed of origin :
 [-8329.06608875 13811.46695562]


 Magnitude of Total Electric Field :
  16128.544924359738


 Direction (angle from x-axis in degrees):
 121.09226324983518


Unit Vector of Total Electric Field:
 [-0.5164177   0.85633683]
```

In [28]:
```
# Q 1) (b)
```

In [30]:
```
import numpy as np

# Moment of inertia tensor (3x3 matrix)
I = np.array([
    [3, 0, 0],
    [0, 5, 0],
    [0, 0, 4]
])


w = np.array([0, 0, 5])


L = I*w


print("Angular Momentum Vector (kg·m²/s):", L)
```

```
Angular Momentum Vector (kg·m²/s): [[ 0  0  0]
 [ 0  0  0]
 [ 0  0 20]]
```

In [ ]:
```
# Q 1) (c)
```

In [31]:
```
import numpy as np

q = 2e-6
v = np.array([3, 4, 0])
B = np.array([0, 0.2, 0.3])


F = q * np.cross(v, B)
```

```
print("Magnetic Force Vector (N):", F)
```

```
Magnetic Force Vector (N): [ 2.4e-06 -1.8e-06  1.2e-06]
```

In [ ]: